ExamGOAL
PYQ PDF

Free · With answer key

JEE MAIN · CHEMISTRY · CHAPTER-WISE PYQS

d and f Block Elements

JEE Main Chemistry · Previous Year Questions with Answer Key

262	160	2002 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

MCQ (Single Correct Answer)	220 questions
Numerical	41 questions
MCQ (More than One Correct Answer)	1 question
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

MCQ (Single Correct Answer) 220 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Open solution

Given below are two statements :

Statement I : The number of pairs among $\left[Ti^{4+}, V^{2+}\right]$, $\left[V^{2+}, Mn^{2+}\right]$, $\left[Mn^{2+}, Fe^{3+}\right]$ and $\left[V^{2+}, Cr^{2+}\right]$ in which both ions are coloured is 3 .

Statement II : The number of pairs among $\left[La^{3+}, Yb^{2+}\right]$, $\left[Lu^{3+}, Ce^{4+}\right]$ and $\left[Ac^{3+}, Lr^{3+}\right]$ ions in which both are diamagnetic is 3 .

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Q2. JEE Main 2026 (Online) 8th April Evening Shift

Open solution

Given below are two statements for catalytic properties of transition metals.

Statement I : First row transition metals which act as catalyst utilise their 3d electrons only for formation of bonds between reactant molecules and atoms on the surface of catalyst.

Statement II : There is increase in the concentration of reactants on the surface of catalyst which strengthens the bonds in reacting molecules.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Q3. JEE Main 2026 (Online) 6th April Evening Shift

Open solution

The correct order of first $(\Delta_i H_1)$ and second $(\Delta_i H_2)$ ionisation enthalpy values of Cr and Mn are :

A. $\Delta_i H_1 : Cr > Mn$

B. $\Delta_i H_2 : Cr > Mn$

C. $\Delta_i H_1 : Mn > Cr$

D. $\Delta_i H_2 : Mn > Cr$

Choose the correct answer from the options given below :

(A) A and B only (B) B and C only (C) A and D only (D) C and D only

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q4. JEE Main 2026 (Online) 6th April Morning Shift

Open solution

The correct statements among the following are :

A. Basic vanadium oxide is used in the manufacture of H_2SO_4.

B. The spin-only magnetic moment value of the transition metal halide employed in Ziegler-Natta polymerization is 2.84 BM .

C. The p-block metal compound employed in Ziegler-Natta polymerization has the metal in +3 oxidation state.

D. The number of electrons present in the outer most ' d ' orbital of metal halide employed in Wacker process is 8.

Choose the correct answer from the options given below :

(A) A and B Only **(B)** A, C and D Only **(C)** C and D Only **(D)** B, C and D Only

Q5. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

A paper dipped in a dil. H_2SO_4 solution of ' X ' upon treatment with SO_2 gas turns into green. The compound ' X ' is :

(A) $KI - strach$ **(B)** $KMnO_4$

(C) $Pb(CH_3COO)_2$ **(D)** $K_2Cr_2O_7$

Q6. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

Which of the following is NOT a physical or chemical characteristics of interstitial compounds?

(A) They have high melting points, higher than those of pure metals.

(B) They are very soft and ionic in nature.

(C) They retain metallic conductivity.

(D) They are chemically inert and usually non-stoichiometric.

Q7. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

Pairs of elements with the same number of electrons in their respective 4f orbital are

[Atomic number. Eu-63, Gd-64, Dy-66, Ho-67, Tm-69, Yb-70, Lu-71, Hf-72]

A. (Eu and Gd)

B. (Dy and Ho)

C. (Yb and Hf)

D. (Lu and Tm)

Choose the correct answer from the options given below :

(A) B and C Only **(B)** A and B Only **(C)** A and D Only **(D)** A and C Only

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q8. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

Consider $|x|$ is the difference in oxidation states of Mn in highest manganese fluoride and highest manganese oxide. The ions with $|x|$ number of unpaired electrons from the following are :

A. Sc^{3+}

B. Zn^{2+}

C. V^{2+}

D. Fe^{2+}

E. Co^{2+}

Choose the correct answer from the options given below :

(A) A and B Only (B) C, D and E Only (C) C and E Only (D) B and E Only

Q9. JEE Main 2026 (Online) 4th April Morning Shift

Open solution

The correct statements among the following are,

A. $Mo(VI)$ and $W(VI)$ are less stable than $Cr(VI)$.

B. Ce^{4+} and Tb^{4+} are oxidant while Eu^{2+} and Yb^{2+} are reductant.

C. Cm and Am have seven unpaired electrons.

D. Actinoid contraction is greater from element to element than lanthanoid contraction.

Choose the correct answer from the options given below :

(A) A and B Only (B) C and D Only (C) B and D Only (D) A and C Only

Q10. JEE Main 2026 (Online) 4th April Morning Shift

Open solution

Correct statements from the following are

A. Potassium dichromate is an oxidising agent and it oxidises $FeSO_4$ to $Fe_2(SO_4)_3$ in acidic medium.

B. Sodium dichromate can be used as primary standard in volumetric estimation.

C. CrO_4^{2-} and $Cr_2O_7^{2-}$ are interconvertible in aqueous solution by varying the pH of the solution.

D. $Cr - O - Cr$ bond angle in $Cr_2O_7^{2-}$ is 126°.

Choose the correct answer from the options given below :

(A) A, B and C Only (B) A, C and D Only (C) A and C Only (D) B and D Only

Q11. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

Among Fe^{2+}, Fe^{3+}, Cr^{2+} and Zn^{2+}, the ion that shows positive borax bead test and with highest ionisation enthalpy is :

(A) Fe^{2+} (B) Zn^{2+} (C) Cr^{2+} (D) Fe^{3+}

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q12. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

Which of the following sets includes all the species that will change the orange colour of $K_2Cr_2O_7$ in acidic medium?

(A) Fe^{2+}, Sn^{2+}, I^-, S^{2-}

(B) S^{2-}, Fe^{3+}, I^-, $C_2O_4^{2-}$

(C) Fe^{2+}, NO_2^-, SO_2, Sn^{4+}

(D) Fe^{3+}, SO_4^{2-}, S^{2-}, Sn^{4+}

Q13. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

Consider the following statements about manganate and permanganate ions. Identify the **correct** statements.

A. The geometry of both manganate and permanganate ions is tetrahedral.

B. The oxidation states of Mn in manganate and permanganate are +7 and +6, respectively.

C. Oxidation of Mn(II) salt by peroxodisulphate gives manganate ion as the final product.

D. Manganate ion is paramagnetic and permanganate ion is diamagnetic.

E. Acidified permanganate ion reduces oxalate, nitrite and iodide ions.

Choose the **correct** answer from the options given below :

(A) A and D Only (B) A, B and C Only (C) A, C and D Only (D) A, D and E Only

Q14. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

Given below are two statements :

Statement I : The number of pairs, from the following, in which both the ions are coloured in aqueous solution is 3.

$[Sc^{3+}, Ti^{3+}]$, $[Mn^{2+}, Cr^{2+}]$, $[Cu^{2+}, Zn^{2+}]$ and $[Ni^{2+}, Ti^{4+}]$

Statement II : Th^{4+} is the strongest reducing agent among Th^{4+}, Ce^{4+}, Gd^{3+} and Eu^{2+}.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Q15. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

" X " is an oxoanion of the lightest element of group 7 (in the periodic table). The metal is in +6 oxidation state in " X ". The color of the potassium salt of X is

(A) purple (B) orange (C) yellow (D) green

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q16. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Given below are two statements :

Statement I : The first ionization enthalpy of Cr is lower than that of Mn .

Statement II : The second and third ionization enthalpies of Cr are higher than those of Mn .

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are true

(C) Both Statement I and Statement II are false

(D) Statement I is false but Statement II is true

Q17. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

On heating a mixture of common salt and $K_2Cr_2O_7$ in equal amount along with concentrated H_2SO_4 in a test tube, a gas is evolved. Formula of the gas evolved and oxidation state of the central metal atom in the gas respectively are:

(A) $Cr_2O_2Cl_2$ and +3 (B) $Cr_2O_2Cl_2$ and +6 (C) CrO_2Cl_2 and +6 (D) CrO_2Cl_2 and +5

Q18. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

Given below are some of the statements about Mn and Mn_2O_7. Identify the correct statements.

A. Mn forms the oxide Mn_2O_7, in which Mn is in its highest oxidation state.

B. Oxygen stabilizes the Mn in higher oxidation states by forming multiple bonds with Mn.

C. Mn_2O_7 is an ionic oxide.

D. The structure of Mn_2O_7 consists of one bridged oxygen.

Choose the **correct** answer from the options given below :

(A) A, B, C and D (B) A, B and D Only (C) A, B and C Only (D) A, C and D Only

Q19. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

MnO_4^{2-}, in acidic medium, disproportionates to :

(A) MnO_4^- and MnO (B) MnO_4^- and MnO_2

(C) Mn_2O_7 and MnO_2 (D) Mn_2O_7 and MnO

Q20. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

The correct decreasing order of spin only magnetic moment values (BM) of Cu^+, Cu^{2+}, Cr^{2+} and Cr^{3+} ions is :

(A) $Cr^{3+} > Cr^{2+} > Cu^+ > Cu^{2+}$ (B) $Cu^+ > Cu^{2+} > Cr^{3+} > Cr^{2+}$

(C) $Cr^{2+} > Cr^{3+} > Cu^{2+} > Cu^+$ (D) $Cu^{2+} > Cu^+ > Cr^{2+} > Cr^{3+}$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q21. JEE Main 2025 (Online) 7th April Morning Shift

The first transition series metal ' M ' has the highest enthalpy of atomisation in its series. One of its aquated ion (M^{n+}) exists in green colour. The nature of the oxide formed by the above M^{n+} ion is:

(A) basic (B) neutral (C) amphoteric (D) acidic

Open solution

Q22. JEE Main 2025 (Online) 7th April Morning Shift

The number of valence electrons present in the metal among $\mathrm{Cr}, \mathrm{Co}, \mathrm{Fe}$ and Ni which has the lowest enthalpy of atomisation is :

(A) 10 (B) 6 (C) 9 (D) 8

Open solution

Q23. JEE Main 2025 (Online) 4th April Morning Shift

Pair of transition metal ions having the same number of unpaired electrons is

(A) $\mathrm{Ti}^{3+}, \mathrm{Mn}^{2+}$ (B) $\mathrm{Ti}^{2+}, \mathrm{Co}^{2+}$

(C) $\mathrm{Fe}^{3+}, \mathrm{Cr}^{2+}$ (D) $\mathrm{V}^{2+}, \mathrm{Co}^{2+}$

Open solution

Q24. JEE Main 2025 (Online) 3rd April Evening Shift

Given below are two statements:

Statement I : CrO_3 is a stronger oxidizing agent than MoO_3

Statement II : $\mathrm{Cr(VI)}$ is more stable than $\mathrm{Mo(VI)}$

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are true

(C) Both Statement I and Statement II are false

(D) Statement I is false but Statement II is true

Open solution

Q25. JEE Main 2025 (Online) 3rd April Morning Shift

The metal ions that have the calculated spin-only magnetic moment value of 4.9 B.M. are :

A. Cr^{2+}

B. Fe^{2+}

C. Fe^{3+}

D. Co^{2+}

E. Mn^{3+}

Choose the correct answer from the options given below:

(A) A, D and E Only (B) A, B and E Only (C) A, C and E Only (D) B and E Only

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q26. JEE Main 2025 (Online) 29th January Morning Shift

The correct option with order of melting points of the pairs (Mn, Fe), (Tc, Ru) and (Re, Os) is :

(A) Fe < Mn, Ru < Tc and Re < Os

(B) Fe < Mn, Ru < Tc and Os < Re

(C) Mn < Fe, Tc < Ru and Re < Os

(D) Mn < Fe, Tc < Ru and Os < Re

Open solution

Q27. JEE Main 2025 (Online) 28th January Evening Shift

The amphoteric oxide among V_2O_3, V_2O_4 and V_2O_5, upon reaction with alkali leads to formation of an oxide anion. The oxidation state of V in the oxide anion is :

(A) + 4 (B) + 5 (C) + 3 (D) + 7

Open solution

Q28. JEE Main 2025 (Online) 28th January Morning Shift

The metal ion whose electronic configuration is not affected by the nature of the ligand and which gives a violet colour in non-luminous flame under hot condition in borax bead test is

(A) Cr^{3+} (B) Ti^{3+} (C) Ni^{2+} (D) Mn^{2+}

Open solution

Q29. JEE Main 2025 (Online) 24th January Evening Shift

Match List - I with List - II.

	List - I (Transition metal ion)		**List - II (Spin only magnetic moment (B.M.))**
(A)	Ti^{3+}	(I)	3.87
(B)	V^{2+}	(II)	0.00
(C)	Ni^{2+}	(III)	1.73
(D)	Sc^{3+}	(IV)	2.84

Choose the correct answer from the options given below :

(A) $(A)-(III),(B)-(I),(C)-(IV),(D)-(II)$

(B) $(A)-(II),(B)-(IV),(C)-(I),(D)-(III)$

(C) $(A)-(III),(B)-(I),(C)-(II),(D)-(IV)$

(D) $(A)-(IV),(B)-(II),(C)-(III),(D)-(I)$

Open solution

Q30. JEE Main 2025 (Online) 24th January Morning Shift

Preparation of potassium permanganate from MnO_2 involves two step process in which the 1st step is a reaction with KOH and KNO_3 to produce

(A) K_3MnO_4 (B) $K_4[Mn(OH)_6]$

(C) K_2MnO_4 (D) $KMnO_4$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q31. JEE Main 2025 (Online) 24th January Morning Shift

Which of the following ions is the strongest oxidizing agent? (Atomic Number of $\mathrm{Ce} = 58, \mathrm{Eu} = 63, \mathrm{Tb} = 65, \mathrm{Lu} = 71$)

(A) Lu^{3+} (B) Eu^{2+} (C) Tb^{4+} (D) Ce^{3+}

Open solution

Q32. JEE Main 2025 (Online) 23rd January Evening Shift

Match List - I with List - II

	List - I		List - II
(A)	Bronze	(I)	Cu, Ni
(B)	Brass	(II)	Fe, Cr, Ni, C
(C)	UK silver coin	(III)	Cu, Zn
(D)	Stainless steel	(IV)	Cu, Sn

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(II), (C)-(III), (D)-(I) (B) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(C) (A)-(III), (B)-(I), (C)-(IV), (D)-(II) (D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Open solution

Q33. JEE Main 2025 (Online) 23rd January Evening Shift

Consider the following reactions

$$\mathrm{K_2Cr_2O_7} \xrightarrow[-\mathrm{H_2O}]{\mathrm{KOH}} [\mathrm{A}] \xrightarrow[-\mathrm{H_2O}]{\mathrm{H_2SO_4}} [\mathrm{B}] + \mathrm{K_2SO_4}$$

The products $[A]$ and $[B]$, respectively are :

(A) $\mathrm{K_2Cr(OH)_6}$ and $\mathrm{Cr_2O_3}$ (B) $\mathrm{K_2CrO_4}$ and $\mathrm{Cr_2O_3}$

(C) $\mathrm{K_2CrO_4}$ and $\mathrm{K_2Cr_2O_7}$ (D) $\mathrm{K_2CrO_4}$ and CrO

Open solution

Q34. JEE Main 2025 (Online) 23rd January Morning Shift

The correct set of ions (aqueous solution) with same colour from the following is:

(A) $\mathrm{V}^{2+}, \mathrm{Cr}^{3+}, \mathrm{Mn}^{3+}$ (B) $\mathrm{Ti}^{4+}, \mathrm{V}^{4+}, \mathrm{Mn}^{2+}$

(C) $\mathrm{Sc}^{3+}, \mathrm{Ti}^{3+}, \mathrm{Cr}^{2+}$ (D) $\mathrm{Zn}^{2+}, \mathrm{V}^{3+}, \mathrm{Fe}^{3+}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q35. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

Lanthanoid ions with $4f^7$ configuration are :

(A) Eu^{2+}

(B) Gd^{3+}

(C) Eu^{3+}

(D) Tb^{3+}

(E) Sm^{2+}

Choose the correct answer from the options given below :

(A) (B) and (E) only **(B)** (B) and (C) only **(C)** (A) and (D) only **(D)** (A) and (B) only

Q36. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

The electronic configuration of Einsteinium is :

(Given atomic number of Einsteinium $= 99$)

(A) $[Rn]5f^{13}6\,d^0 7\,s^2$ **(B)** $[Rn]5f^{12}6\,d^0 7\,s^2$

(C) $[Rn]5f^{11}6\,d^0 7\,s^2$ **(D)** $[Rn]5f^{10}6\,d^0 7\,s^2$

Q37. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

Given below are two statements :

Statement I : The higher oxidation states are more stable down the group among transition elements unlike p-block elements.

Statement II : Copper can not liberate hydrogen from weak acids.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are false

(C) Both Statement I and Statement II are true

(D) Statement I is false but Statement II is true

Q38. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

The electronic configuration of $Cu(II)$ is $3\,d^9$ whereas that of $Cu(I)$ is $3\,d^{10}$. Which of the following is correct?

(A) $Cu(II)$ is more stable

(B) Cu (II) is less stable

(C) $Cu(I)$ and $Cu(II)$ are equally stable

(D) Stability of $Cu(I)$ and $Cu(II)$ depends on nature of copper salts

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q39. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

Given below are two statements :

Statement (I) : Fusion of MnO_2 with KOH and an oxidising agent gives dark green K_2MnO_4.

Statement (II) : Manganate ion on electrolytic oxidation in alkaline medium gives permanganate ion.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

Q40. JEE Main 2024 (Online) 8th April Morning Shift

Open solution

Iron (III) catalyses the reaction between iodide and persulphate ions, in which

A. Fe^{3+} oxidises the iodide ion

B. Fe^{3+} oxidises the persulphate ion

C. Fe^{2+} reduces the iodide ion

D. Fe^{2+} reduces the persulphate ion

Choose the most appropriate answer from the options given below:

(A) B and C only **(B)** A only **(C)** A and D only **(D)** B only

Q41. JEE Main 2024 (Online) 6th April Evening Shift

Open solution

Arrange the following elements in the increasing order of number of unpaired electrons in it.

(A) Sc

(B) Cr

(C) V

(D) Ti

(E) Mn

Choose the correct answer from the options given below :

(A) $(A) < (D) < (C) < (B) < (E)$ **(B)** $(A) < (D) < (C) < (E) < (B)$

(C) $(B) < (C) < (D) < (E) < (A)$ **(D)** $(C) < (E) < (B) < (A) < (D)$

Q42. JEE Main 2024 (Online) 6th April Morning Shift

Open solution

The number of elements from the following that do not belong to lanthanoids is Eu, Cm, Er, Tb, Yb and Lu

(A) 3 **(B)** 1 **(C)** 4 **(D)** 5

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q43. JEE Main 2024 (Online) 5th April Evening Shift

While preparing crystals of Mohr's salt, dil H_2SO_4 is added to a mixture of ferrous sulphate and ammonium sulphate, before dissolving this mixture in water, dil H_2SO_4 is added here to :

(A) prevent the hydrolysis of ferrous sulphate

(B) prevent the hydrolysis of ammonium sulphate

(C) increase the rate of formation of crystals

(D) make the medium strongly acidic

Open solution

Q44. JEE Main 2024 (Online) 5th April Evening Shift

The number of ions from the following that have the ability to liberate hydrogen from a dilute acid is _____.

Ti^{2+}, Cr^{2+} and V^{2+}

(A) 1 (B) 2 (C) 3 (D) 0

Open solution

Q45. JEE Main 2024 (Online) 5th April Morning Shift

The metal that shows highest and maximum number of oxidation state is :

(A) Co (B) Mn (C) Fe (D) Ti

Open solution

Q46. JEE Main 2024 (Online) 4th April Morning Shift

The element which shows only one oxidation state other than its elemental form is :

(A) Nickel (B) Titanium (C) Cobalt (D) Scandium

Open solution

Q47. JEE Main 2024 (Online) 1st February Evening Shift

Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In aqueous solutions Cr^{2+} is reducing while Mn^{3+} is oxidising in nature.

Reason (R) : Extra stability to half filled electronic configuration is observed than incompletely filled electronic configuration.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(A) (A) is true but (R) is false

(B) Both (A) and (R) are true and (R) is the correct explanation of (A)

(C) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(D) (A) is false but (R) is true

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q48. JEE Main 2024 (Online) 31st January Evening Shift

Open solution

Choose the correct statements from the following

A. Mn_2O_7 is an oil at room temperature

B. V_2O_4 reacts with acid to give VO_2^{2+}

C. CrO is a basic oxide

D. V_2O_5 does not react with acid

Choose the correct answer from the options given below :

(A) A, B and D only **(B)** A, B and C only **(C)** A and C only **(D)** B and C only

Q49. JEE Main 2024 (Online) 31st January Morning Shift

Open solution

Identify correct statements from below:

A. The chromate ion is square planar.

B. Dichromates are generally prepared from chromates.

C. The green manganate ion is diamagnetic.

D. Dark green coloured K_2MnO_4 disproportionates in a neutral or acidic medium to give permanganate.

E. With increasing oxidation number of transition metal, ionic character of the oxides decreases.

Choose the correct answer from the options given below:

(A) B, D, E only **(B)** A, B, C only **(C)** A, D, E only **(D)** B, C, D only

Q50. JEE Main 2024 (Online) 31st January Morning Shift

Open solution

The metals that are employed in the battery industries are

A. Fe

B. Mn

C. Ni

D. Cr

E. Cd

Choose the correct answer from the options given below:

(A) A, B, C, D and E **(B)** A, B, C and D only

(C) B, D and E only **(D)** B, C and E only

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q51. JEE Main 2024 (Online) 30th January Evening Shift

Open solution

A and B formed in the following reactions are:

$CrO_2Cl_2 + 4NaOH \rightarrow A + 2NaCl + 2H_2O$,

$A + 2HCl + 2H_2O_2 \rightarrow B + 3H_2O$

(A) $A = Na_2Cr_2O_7,\ B = CrO_5$

(B) $A = Na_2CrO_4,\ B = CrO_5$

(C) $A = Na_2Cr_2O_4,\ B = CrO_4$

(D) $A = Na_2Cr_2O_7,\ B = CrO_3$

Q52. JEE Main 2024 (Online) 30th January Evening Shift

Open solution

Alkaline oxidative fusion of MnO_2 gives "A" which on electrolytic oxidation in alkaline solution produces B. A and B respectively are

(A) Mn_2O_3 and MnO_4^{2-}

(B) Mn_2O_7 and MnO_4^-

(C) MnO_4^{2-} and MnO_4^-

(D) MnO_4^{2-} and Mn_2O_7

Q53. JEE Main 2024 (Online) 30th January Evening Shift

Open solution

The orange colour of $K_2Cr_2O_7$ and purple colour of $KMnO_4$ is due to

(A) Charge transfer transition in both.

(B) $d \rightarrow d$ transitions in $KMnO_4$ and charge transfer transitions in $K_2Cr_2O_7$.

(C) $d \rightarrow d$ transitions in both

(D) $d \rightarrow d$ transitions in $K_2Cr_2O_7$ and charge transfer transitions in $KMnO_4$.

Q54. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

Match List I with List II.

	List I Species		List II Electronic distribution
(A)	Cr^{+2}	(I)	$3d^8$
(B)	Mn^+	(II)	$3d^34s^1$
(C)	Ni^{+2}	(III)	$3d^4$
(D)	V^+	(IV)	$3d^54s^1$

Choose the correct answer from the options given below :

(A) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(B) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(C) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(D) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

Q55. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

Diamagnetic Lanthanoid ions are :

(A) La^{3+} & Ce^{4+}

(B) Nd^{3+} & Ce^{4+}

(C) Lu^{3+} & Eu^{3+}

(D) Nd^{3+} & Eu^{3+}

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q56. JEE Main 2024 (Online) 29th January Evening Shift

Open solution

Which of the following acts as a strong reducing agent? (Atomic number: $\mathrm{Ce} = 58, \mathrm{Eu} = 63, \mathrm{Gd} = 64, \mathrm{Lu} = 71$)

(A) Eu^{2+} (B) Gd^{3+} (C) Lu^{3+} (D) Ce^{4+}

Q57. JEE Main 2024 (Online) 29th January Evening Shift

Open solution

Which of the following statements are correct about Zn, Cd and Hg ?

A. They exhibit high enthalpy of atomization as the d-subshell is full.

B. Zn and Cd do not show variable oxidation state while Hg shows $+\mathrm{I}$ and $+\mathrm{II}$.

C. Compounds of Zn, Cd and Hg are paramagnetic in nature.

D. Zn, Cd and Hg are called soft metals.

Choose the most appropriate from the options given below:

(A) C, D only (B) B, C only (C) A, D only (D) B, D only

Q58. JEE Main 2024 (Online) 29th January Evening Shift

Open solution

The correct IUPAC name of $\mathrm{K_2MnO_4}$ is

(A) Potassium tetraoxidomanganate (VI)

(B) Dipotassium tetraoxidomanganate (VII)

(C) Potassium tetraoxopermanganate (VI)

(D) Potassium tetraoxidomanganese (VI)

Q59. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

$\mathrm{KMnO_4}$ decomposes on heating at $513\ \mathrm{K}$ to form $\mathrm{O_2}$ along with

(A) $\mathrm{K_2MnO_4}$ & Mn

(B) $\mathrm{MnO_2}$ & $\mathrm{K_2O_2}$

(C) $\mathrm{K_2MnO_4}$ & $\mathrm{MnO_2}$

(D) Mn & $\mathrm{KO_2}$

Q60. JEE Main 2024 (Online) 27th January Evening Shift

Open solution

Given below are two statements :

Statement (I) : In the Lanthanoids, the formation of Ce^{+4} is favoured by its noble gas configuration.

Statement (II) : Ce^{+4} is a strong oxidant reverting to the common +3 state.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are true

(D) Statement I is false but Statement II is true

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q61. JEE Main 2024 (Online) 27th January Evening Shift

Choose the correct option having all the elements with d^{10} electronic configuration from the following :

(A) $^{46}Pd, ^{28}Ni, ^{26}Fe, ^{24}Cr$

(B) $^{29}Cu, ^{30}Zn, ^{48}Cd, ^{47}Ag$

(C) $^{27}Co, ^{28}Ni, ^{26}Fe, ^{24}Cr$

(D) $^{28}Ni, ^{24}Cr, ^{26}Fe, ^{29}Cu$

Open solution

Q62. JEE Main 2024 (Online) 27th January Morning Shift

Which of the following electronic configuration would be associated with the highest magnetic moment?

(A) $[Ar]3\,d^7$ (B) $[Ar]3\,d^8$ (C) $[Ar]3\,d^3$ (D) $[Ar]3\,d^6$

Open solution

Q63. JEE Main 2024 (Online) 27th January Morning Shift

$NaCl$ reacts with conc. H_2SO_4 and $K_2Cr_2O_7$ to give reddish fumes (B), which react with $NaOH$ to give yellow solution (C). (B) and (C) respectively are ;

(A) CrO_2Cl_2, Na_2CrO_4 (B) Na_2CrO_4, CrO_2Cl_2

(C) $CrO_2Cl_2, KHSO_4$ (D) $CrO_2Cl_2, Na_2Cr_2O_7$

Open solution

Q64. JEE Main 2024 (Online) 27th January Morning Shift

The electronic configuration for Neodymium is:

[Atomic Number for Neodymium 60]

(A) $[Xe]4f^4 6\,s^2$ (B) $[Xe]5f^4 7\,s^2$

(C) $[Xe]4f^6 6\,s^2$ (D) $[Xe]4f^1 5\,d^1 6\,s^2$

Open solution

Q65. JEE Main 2023 (Online) 13th April Morning Shift

The pair of lanthanides in which both elements have high third - ionization energy is :

(A) Dy, Gd (B) Lu, Yb (C) Eu, Yb (D) Eu, Gd

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q66. JEE Main 2023 (Online) 12th April Morning Shift

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A: $5\mathrm{f}$ electrons can participate in bonding to a far greater extent than $4\mathrm{f}$ electrons

Reason R: $5\mathrm{f}$ orbitals are not as buried as $4\mathrm{f}$ orbitals

In the light of the above statements, choose the correct answer from the options given below

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

Q67. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

Prolonged heating is avoided during the preparation of ferrous ammonium sulphate to :

(A) prevent hydrolysis

(B) prevent breaking

(C) prevent oxidation

(D) prevent reduction

Q68. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

Which of the following statements are correct?

(A) The M^{3+}/M^{2+} reduction potential for iron is greater than manganese.

(B) The higher oxidation states of first row d-block elements get stabilized by oxide ion.

(C) Aqueous solution of Cr^{2+} can liberate hydrogen from dilute acid.

(D) Magnetic moment of V^{2+} is observed between 4.4 - 5.2 BM.

Choose the correct answer from the options given below :

(A) (B), (C) only

(B) (A), (B) only

(C) (A), (B), (D) only

(D) (C), (D) only

Q69. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

Which halogen is known to cause the reaction given below :

$2Cu^{2+} + 4X^{-} \rightarrow Cu_2X_2(s) + X_2$

(A) All halogens

(B) Only Iodine

(C) Only Bromine

(D) Only Chlorine

Q70. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

In chromyl chloride, the number of d-electrons present on chromium is same as in

(Given at no. of $\mathrm{Ti:22,\ V:23, Cr:24, Mn:25, Fe:26}$)

(A) Mn (VII)

(B) Ti (III)

(C) Fe (III)

(D) V (IV)

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q71. JEE Main 2023 (Online) 6th April Evening Shift

Element not present in Nessler's reagent is :

(A) I (B) N (C) K (D) Hg

Open solution

Q72. JEE Main 2023 (Online) 6th April Evening Shift

Given below are two statements: one is labelled as "Assertion A" and the other is labelled as "Reason R"

Assertion A : In the complex $\mathrm{Ni(CO)_4}$ and $\mathrm{Fe(CO)_5}$, the metals have zero oxidation state.

Reason R : Low oxidation states are found when a complex has ligands capable of π-donor character in addition to the σ-bonding.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) A is not correct but R is correct

(B) A is correct but R is not correct

(C) Both A and R are correct but R is NOT the correct explanation of A

(D) Both A and R are correct and R is the correct explanation of A

Open solution

Q73. JEE Main 2023 (Online) 6th April Evening Shift

During the reaction of permanganate with thiosulphate, the change in oxidation of manganese occurs by value of 3. Identify which of the below medium will favour the reaction.

(A) aqueous neutral (B) both aqueous acidic and faintly alkaline

(C) both aqueous acidic and neutral (D) aqueous acidic

Open solution

Q74. JEE Main 2023 (Online) 6th April Morning Shift

Strong reducing and oxidizing agents among the following, respectively, are :

(A) $\mathrm{Ce^{4+}}$ and $\mathrm{Eu^{2+}}$ (B) $\mathrm{Eu^{2+}}$ and $\mathrm{Ce^{4+}}$

(C) $\mathrm{Ce^{4+}}$ and $\mathrm{Tb^{4+}}$ (D) $\mathrm{Ce^{3+}}$ and $\mathrm{Ce^{4+}}$

Open solution

Q75. JEE Main 2023 (Online) 1st February Evening Shift

Which element is not present in Nessler's reagent?

(A) Mercury (B) Iodine (C) Potassium (D) Oxygen

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q76. JEE Main 2023 (Online) 1st February Evening Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Cu^{2+} in water is more stable than Cu^{+}.

Reason (R) : Enthalpy of hydration for Cu^{2+} is much less than that of Cu^{+}.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is correct but (R) is not correct

(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(C) (A) is not correct but (R) is correct

(D) Both (A) and (R) are correct and (R) is the correct explanation of (A)

Q77. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

Highest oxidation state of Mn is exhibited in Mn_2O_7. The correct statements about Mn_2O_7 are

(A) Mn is tetrahedrally surrounded by oxygen atoms.

(B) Mn is octahedrally surrounded by oxygen atoms.

(C) Contains Mn-O-Mn bridge.

(D) Contains Mn-Mn bond.

Choose the correct answer from the options given below :

(A) A and C only **(B)** A and D only **(C)** B and C only **(D)** B and D only

Q78. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

Which of the following elements have half-filled f-orbitals in their ground state?

(Given : atomic number $Sm = 62; Eu = 63; Tb = 65; Gd = 64, Pm = 61$)

A. Sm

B. Eu

C. Tb

D. Gd

E. Pm

Choose the correct answer from the options given below :

(A) B and D only **(B)** A and B only **(C)** C and D only **(D)** A and E only

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q79. JEE Main 2023 (Online) 31st January Morning Shift

When Cu^{2+} ion is treated with KI, a white precipitate, X appears in solution. The solution is titrated with sodium thiosulphate, the compound Y is formed. X and Y respectively are :

(A) $X = CuI_2$ | $Y = Na_2S_4O_6$

(B) $X = Cu_2I_2$ | $Y = Na_2S_4O_6$

(C) $X = CuI_2$ | $Y = Na_2S_2O_3$

(D) $X = Cu_2I_2$ | $Y = Na_2S_4O_5$

Open solution

Q80. JEE Main 2023 (Online) 31st January Morning Shift

Nd^{2+} = _____

(A) $4f^4 6s^2$ (B) $4f^3$ (C) $4f^4$ (D) $4f^2 6s^2$

Open solution

Q81. JEE Main 2023 (Online) 31st January Morning Shift

The correct order of basicity of oxides of vanadium is :

(A) $V_2O_3 > V_2O_5 > V_2O_4$ (B) $V_2O_3 > V_2O_4 > V_2O_5$

(C) $V_2O_5 > V_2O_4 > V_2O_3$ (D) $V_2O_4 > V_2O_3 > V_2O_5$

Open solution

Q82. JEE Main 2023 (Online) 30th January Evening Shift

Formulae for Nessler's reagent is :

(A) $KHgI_3$ (B) HgI_2 (C) KHg_2I_2 (D) K_2HgI_4

Open solution

Q83. JEE Main 2023 (Online) 30th January Evening Shift

$KMnO_4$ oxidises I^- in acidic and neutral/faintly alkaline solutions, respectively, to :

(A) IO_3^- & IO_3^- (B) I_2 & I_2

(C) I_2 & IO_3^- (D) IO_3^- & I_2

Open solution

Q84. JEE Main 2023 (Online) 30th January Morning Shift

During the qualitative analysis of SO_3^{2-} using dilute H_2SO_4, SO_2 gas is evolved which turns $K_2Cr_2O_7$ solution (acidified with dilute H_2SO_4) :

(A) blue (B) black (C) red (D) green

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q85. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

Given below are two statements :

Statement I : Nickel is being used as the catalyst for producing syn gas and edible fats.

Statement II : Silicon forms both electron rich and electron deficient hydrides.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but statement II is incorrect

(B) Both the statements I and II are correct

(C) Both the statements I and II are incorrect

(D) Statement I is incorrect but statement II is correct

Q86. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

A solution of CrO_5 in amyl alcohol has a _____ colour.

(A) Yellow (B) Orange-Red (C) Green (D) Blue

Q87. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

The set of correct statements is :

(i) Manganese exhibits +7 oxidation state in its oxide.

(ii) Ruthenium and Osmium exhibit +8 oxidation state in their oxides.

(iii) Sc shows +4 oxidation state which is oxidizing in nature.

(iv) Cr shows oxidising nature in +6 oxidation state.

(A) (ii), (iii) and (iv) (B) (ii) and (iii)

(C) (i) and (iii) (D) (i), (ii) and (iv)

Q88. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

A chloride salt solution acidified with dil.HNO_3 gives a curdy white precipitate, [A], on addition of $AgNO_3$. [A] on treatment with NH_4OH gives a clear solution B. A and B are respectively :

(A) $H[AgCl_3]$ & $(NH_4)[Ag(OH)_2]$ (B) $AgCl$ & $[Ag(NH_3)_2]Cl$

(C) $H[AgCl_3]$ & $[Ag(NH_3)_2]Cl$ (D) $AgCl$ & $(NH_4)[Ag(OH)_2]$

Q89. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

Potassium dichromate acts as a strong oxidizing agent in acidic solution. During this process, the oxidation state changes from :

(A) + 2 to + 1 (B) + 6 to + 2 (C) + 6 to + 3 (D) + 3 to + 1

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q90. JEE Main 2023 (Online) 24th January Evening Shift

Which one amongst the following are good oxidizing agents?

A. Sm^{2+}

B. Ce^{2+}

C. Ce^{4+}

D. Tb^{4+}

Choose the most appropriate answer from the options given below :

(A) C and D only (B) A and B only (C) D only (D) C only

Open solution

Q91. JEE Main 2023 (Online) 24th January Morning Shift

An ammoniacal metal salt solution gives a brilliant red precipitate on addition of dimethylglyoxime. The metal ion is :

(A) Co^{2+} (B) Fe^{2+} (C) Ni^{2+} (D) Cu^{2+}

Open solution

Q92. JEE Main 2022 (Online) 29th July Evening Shift

Which of the following 3d-metal ion will give the lowest enthalpy of hydration $(\Delta_{hyd}H)$ when dissolved in water ?

(A) Cr^{2+} (B) Mn^{2+} (C) Fe^{2+} (D) Co^{2+}

Open solution

Q93. JEE Main 2022 (Online) 29th July Morning Shift

The reaction of zinc with excess of aqueous alkali, evolves hydrogen gas and gives :

(A) $Zn(OH)_2$ (B) ZnO

(C) $[Zn(OH)_4]^{2-}$ (D) $[ZnO_2]^{2-}$

Open solution

Q94. JEE Main 2022 (Online) 29th July Morning Shift

In following pairs, the one in which both transition metal ions are colourless is :

(A) Sc^{3+}, Zn^{2+} (B) Ti^{4+}, Cu^{2+}

(C) V^{2+}, Ti^{3+} (D) Zn^{2+}, Mn^{2+}

Open solution

Q95. JEE Main 2022 (Online) 29th July Morning Shift

In neutral or faintly alkaline medium, $KMnO_4$ being a powerful oxidant can oxidize, thiosulphate almost quantitatively, to sulphate. In this reaction overall change in oxidation state of manganese will be :

(A) 5 (B) 1 (C) 0 (D) 3

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q96. JEE Main 2022 (Online) 28th July Morning Shift

Open solution

Match List - I with List - II, match the gas evolved during each reaction.

	List - I		List - II
(A)	$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$	(I)	H_2
(B)	$KMnO_4 + HCl \rightarrow$	(II)	N_2
(C)	$Al + NaOH + H_2O \rightarrow$	(III)	O_2
(D)	$NaNO_3 \xrightarrow{\Delta}$	(IV)	Cl_2

Choose the correct answer from the options given below :

(A) $(A) - (II), (B) - (III), (C) - (I), (D) - (IV)$

(B) $(A) - (III), (B) - (I), (C) - (IV), (D) - (II)$

(C) $(A) - (II), (B) - (IV), (C) - (I), (D) - (III)$

(D) $(A) - (III), (B) - (IV), (C) - (I), (D) - (II)$

Q97. JEE Main 2022 (Online) 28th July Morning Shift

Open solution

Which of the following has least tendency to liberate H_2 from mineral acids?

(A) Cu **(B)** Mn **(C)** Ni **(D)** Zn

Q98. JEE Main 2022 (Online) 27th July Evening Shift

Open solution

In neutral or alkaline solution, MnO_4^- oxidises thiosulphate to :

(A) $S_2O_7^{2-}$ **(B)** $S_2O_8^{2-}$

(C) SO_3^{2-} **(D)** SO_4^{2-}

Q99. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

Given below are two statements.

Statement I : Iron (III) catalyst, acidified $K_2Cr_2O_7$ and neutral $KMnO_4$ have the ability to oxidise I^- to I_2 independently.

Statement II : Manganate ion is paramagnetic in nature and involves $p\pi - p\pi$ bonding.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q100. JEE Main 2022 (Online) 27th July Morning Shift

The total number of $Mn = O$ bonds in Mn_2O_7 is _____.

(A) 4 (B) 5 (C) 6 (D) 3

Open solution

Q101. JEE Main 2022 (Online) 26th July Morning Shift

The dark purple colour of $KMnO_4$ disappears in the titration with oxalic acid in acidic medium. The overall change in the oxidation number of manganese in the reaction is :

(A) 5 (B) 1 (C) 7 (D) 2

Open solution

Q102. JEE Main 2022 (Online) 29th June Morning Shift

The electronic configuration of Pt (atomic number 78) is :

(A) [Xe] $4f^{14}$ $5d^9$ $6s^1$ (B) [Kr] $4f^{14}$ $5d^{10}$

(C) [Xe] $4f^{14}$ $5d^{10}$ (D) [Xe] $4f^{14}$ $5d^8$ $6s^2$

Open solution

Q103. JEE Main 2022 (Online) 28th June Morning Shift

Which one of the lanthanoids given below is the most stable in divalent form?

(A) Ce (Atomic Number 58) (B) Sm (Atomic number 62)

(C) Eu (Atomic Number 63) (D) Yb (Atomic Number 70)

Open solution

Q104. JEE Main 2022 (Online) 27th June Evening Shift

In 3d series, the metal having the highest M^{2+}/M standard electrode potential is :

(A) Cr (B) Fe (C) Cu (D) Zn

Open solution

Q105. JEE Main 2022 (Online) 27th June Evening Shift

The 'f' orbitals are half and completely filled, respectively in lanthanide ions :

[Given : Atomic no. Eu, 63; Sm, 62; Tm, 69; Tb, 65; Yb, 70; Dy, 66]

(A) Eu^{2+} and Tm^{2+} (B) Sm^{2+} and Tm^{3+} (C) Tb^{4+} and Yb^{2+} (D) Dy^{3+} and Yb^{3+}

Open solution

Q106. JEE Main 2022 (Online) 26th June Evening Shift

The most common oxidation state of Lanthanoid elements is +3. Which of the following is likely to deviate easily from +3 oxidation state?

(A) Ce (At. No. 58) (B) La (At. No. 57) (C) Lu (At. No. 71) (D) Gd (At. No. 64)

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q107. JEE Main 2022 (Online) 25th June Evening Shift

The metal ion (in gaseous state) with lowest spin-only magnetic moment value is :

(A) V^{2+} (B) Ni^{2+} (C) Cr^{2+} (D) Fe^{2+}

Open solution

Q108. JEE Main 2022 (Online) 25th June Morning Shift

Cerium (IV) has a noble gas configuration. Which of the following is correct statement about it?

(A) It will not prefer to undergo redox reactions.

(B) It will prefer to gain electron and act as an oxidizing agent.

(C) It will prefer to give away an electron and behave as reducing agent.

(D) It acts as both, oxidizing and reducing agent.

Open solution

Q109. JEE Main 2022 (Online) 25th June Morning Shift

Among the following, which is the strongest oxidizing agent?

(A) Mn^{3+} (B) Fe^{3+} (C) Ti^{3+} (D) Cr^{3+}

Open solution

Q110. JEE Main 2022 (Online) 24th June Evening Shift

Metals generally melt at very high temperature. Amongst the following, the metal with the highest melting point will be :

(A) Hg (B) Ag (C) Ga (D) Cs

Open solution

Q111. JEE Main 2021 (Online) 1st September Evening Shift

In the given chemical reaction, colors of the Fe^{2+} and Fe^{3+} ions, are respectively :

$$5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$$

(A) Yellow, Orange (B) Yellow, Green (C) Green, Orange (D) Green, Yellow

Open solution

Q112. JEE Main 2021 (Online) 1st September Evening Shift

Identify the element for which electronic configuration in +3 oxidation state is $[Ar]3d^5$:

(A) Ru (B) Mn (C) Co (D) Fe

Open solution

Q113. JEE Main 2021 (Online) 31st August Evening Shift

The Eu^{2+} ion is a strong reducing agent in spite of its ground state electronic configuration (outermost) : [Atomic number of Eu = 63]

(A) $4f^7 6s^2$ (B) $4f^6$ (C) $4f^7$ (D) $4f^6 6s^2$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q114. JEE Main 2021 (Online) 31st August Evening Shift

In which one of the following sets all species show disproportionation reaction?

(A) ClO_2^-, F_2, MnO_4^- and $Cr_2O_7^{2-}$

(B) $Cr_2O_7^{2-}$, MnO_4^-, ClO_2^- and Cl_2

(C) MnO_4^-, ClO_2^-, Cl_2 and Mn^{3+}

(D) ClO_4^-, MnO_4^-, ClO_2^- and F_2

Open solution

Q115. JEE Main 2021 (Online) 31st August Morning Shift

In the structure of the dichromate ion, there is a :

(A) linear symmetrical Cr-O-Cr bond.

(B) non-linear symmetrical Cr-O-Cr bond.

(C) linear unsymmetrical Cr-O-Cr bond.

(D) non-linear unsymmetrical Cr-O-Cr bond.

Open solution

Q116. JEE Main 2021 (Online) 31st August Morning Shift

Which one of the following lanthanides exhibits +2 oxidation state with diamagnetic nature? (Given Z for Nd = 60, Yb = 70, La = 57, Ce = 58)

(A) Nd (B) Yb (C) La (D) Ce

Open solution

Q117. JEE Main 2021 (Online) 27th August Evening Shift

The addition of dilute NaOH to Cr^{3+} salt solution will give :

(A) a solution of $[Cr(OH)_4]^-$

(B) precipitate of $Cr_2O_3(H_2O)_n$

(C) precipitate of $[Cr(OH)_6]^{3-}$

(D) precipitate of $Cr(OH)_3$

Open solution

Q118. JEE Main 2021 (Online) 27th August Evening Shift

Potassium permanganate on heating at 513 K gives a product which is :

(A) paramagnetic and colourless

(B) diamagnetic and green

(C) diamagnetic and colourless

(D) paramagnetic and green

Open solution

Q119. JEE Main 2021 (Online) 27th August Morning Shift

The nature of oxides V_2O_3 and CrO is indexed as 'X' and 'Y' type respectively. The correct set of X and Y is :

(A) X = basic | Y = amphoteric

(B) X = amphoteric | Y = basic

(C) X = acidic | Y = acidic

(D) X = basic | Y = basic

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q120. JEE Main 2021 (Online) 26th August Morning Shift

Which one of the following when dissolved in water gives coloured solution in nitrogen atmosphere?

(A) $CuCl_2$ (B) AgCl (C) $ZnCl_2$ (D) Cu_2Cl_2

Open solution

Q121. JEE Main 2021 (Online) 25th July Evening Shift

The spin only magnetic moments (in BM) for free Ti^{3+}, V^{2+} and Sc^{3+} ions respectively are :

(At.No. Sc : 21, Ti : 22, V : 23)

(A) 3.87, 1.73, 0 (B) 1.73, 3.87, 0 (C) 1.73, 0, 3.87 (D) 0, 3.87, 1.73

Open solution

Q122. JEE Main 2021 (Online) 25th July Evening Shift

Which one of the following metals forms interstitial hydride easily?

(A) Cr (B) Fe (C) Mn (D) Co

Open solution

Q123. JEE Main 2021 (Online) 25th July Morning Shift

The correct order of following 3d metal oxides, according to their oxidation numbers is :

(a) CrO_3, (b) Fe_2O_3, (c) MnO_2, (d) V_2O_5, (e) Cu_2O

(A) (d) > (a) > (b) > (c) > (e) (B) (a) > (c) > (d) > (b) > (e)
(C) (a) > (d) > (c) > (b) > (e) (D) (c) > (a) > (d) > (e) > (b)

Open solution

Q124. JEE Main 2021 (Online) 22th July Evening Shift

The set having ions which are coloured and paramagnetic both is :

(A) Cu^{2+}, Cr^{3+}, Sc^{+} (B) Cu^{2+}, Zn^{2+}, Mn^{4+} (C) Sc^{3+}, V^{5+}, Ti^{4+} (D) Ni^{2+}, Mn^{7+}, Hg^{2+}

Open solution

Q125. JEE Main 2021 (Online) 18th March Evening Shift

The oxide that shows magnetic property is :

(A) MgO (B) SiO_2 (C) Mn_3O_4 (D) Na_2O

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q126. JEE Main 2021 (Online) 18th March Morning Shift

Match List - I with List - II

Open solution

	List - I		List - II
(a)	Chlorophyll	(i)	Ruthenium
(b)	Vitamin-B_{12}	(ii)	Platinum
(c)	Anticancer drug	(iii)	Cobalt
(d)	Grubbs catalyst	(iv)	Magnesium

Choose the most appropriate answer from the options given below :

(A) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

(B) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

(C) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

(D) (a) - (iv), (b) - (ii), (c) - (iii), (d) - (i)

Q127. JEE Main 2021 (Online) 17th March Evening Shift

The common positive oxidation states for an element with atomic number 24, are :

(A) +1 and +3 **(B)** +2 to +6 **(C)** +1 to +6 **(D)** +1 and +3 to +6

Open solution

Q128. JEE Main 2021 (Online) 17th March Morning Shift

What is the spin-only magnetic moment value (BM) of a divalent metal ion with atomic number 25, in it's aqueous solution?

(A) zero **(B)** 5.26 **(C)** 5.0 **(D)** 5.92

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q129. JEE Main 2021 (Online) 17th March Morning Shift

Open solution

Given below are two statements :

Statement I : Potassium permanganate on heating at 573K forms potassium manganate.

Statement II : Both potassium permanganate and potassium manganate are tetrahedral and paramagnetic in nature.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is true but statement II is false

(B) Statement I is false but statement II is true

(C) Both statement I and statement II are false

(D) Both statement I and statement II are true

Q130. JEE Main 2021 (Online) 16th March Evening Shift

Open solution

Fex_2 and Fey_3 are known when x and y are :

(A) x = F, Cl, Br, I and y = F, Cl, Br, I

(B) x = F, Cl, Br and y = F, Cl, Br, I

(C) x = F, Cl, Br, I and y = F, Cl, Br

(D) x = Cl, Br, I and y = F, Cl, Br, I

Q131. JEE Main 2021 (Online) 16th March Evening Shift

Open solution

Arrange the following metal complex/compounds in the increasing order of spin only magnetic moment. Presume all the three, high spin system.

(Atomic numbers Ce = 58, Gd = 64 and Eu = 63)

(a) $(NH_4)_2[Ce(NO_3)_6]$

(b) $Gd(NO_3)_3$ and

(c) $Eu(NO_3)_3$

(A) (a) < (b) < (c)

(B) (b) < (a) < (c)

(C) (a) < (c) < (b)

(D) (c) < (a) < (b)

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q132. JEE Main 2021 (Online) 16th March Morning Shift

Open solution

Given below are two statements :

Statement I : The E^{o} value for Ce^{4+} / Ce^{3+} is + 1.74 V.

Statement II : Ce is more stable in Ce^{4+} state than Ce^{3+} state.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is incorrect but statement II is correct

(B) Both statement I and statement II are correct

(C) Both statement I and statement II are incorrect

(D) Statement I is correct but statement II is incorrect

Q133. JEE Main 2021 (Online) 16th March Morning Shift

Open solution

Given below are two statement : one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : Size of Bk^{3+} ion is less than Np^{3+} ion.

Reason R : The above is a consequence of the lanthanoid contraction.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both A and B are true and R is the correct explanation of A

(B) A is false but R is true

(C) A is true but R is false

(D) Both A and B are true but R is not the correct explanation of A

Q134. JEE Main 2021 (Online) 26th February Morning Shift

Open solution

Which one of the following lanthanoids does not form MO_2? [M is lanthanoid metal]

(A) Nd (B) Dy (C) Yb (D) Pr

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q135. JEE Main 2021 (Online) 25th February Morning Shift

Given below are two statements :

Statement I : CeO_2 can be used for oxidation of aldehydes and ketones.

Statement II : Aqueous solution of $EuSO_4$ is a strong reducing agent.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false.

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

Open solution

Q136. JEE Main 2021 (Online) 25th February Morning Shift

In which of the following pairs, the outer most electronic configuration will be the same?

(A) Cr^+ and Mn^{2+} (B) Ni^{2+} and Cu^+ (C) V^{2+} and Cr^+ (D) Fe^{2+} and Co^+

Open solution

Q137. JEE Main 2021 (Online) 24th February Evening Shift

The **incorrect** statement among the following is :

(A) RuO_4 is an oxidizing agent

(B) Cr_2O_3 is an amphoteric oxide.

(C) $VOSO_4$ is a reducing agent

(D) Red colour of ruby is due to the presence of Co^{3+}

Open solution

Q138. JEE Main 2021 (Online) 24th February Evening Shift

What is the correct order of the following elements with respect to their density?

(A) Cr < Zn < Co < Cu < Fe (B) Cr < Fe < Co < Cu < Zn

(C) Zn < Cr < Fe < Co < Cu (D) Zn < Cu < Co < Fe < Cr

Open solution

Q139. JEE Main 2021 (Online) 24th February Morning Shift

The electrode potential of M^{2+}/M of 3d-series elements shows positive value for :

(A) Zn (B) Fe (C) Cu (D) Co

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q140. JEE Main 2021 (Online) 24th February Morning Shift

Given below are two statements :

Statement I : Colourless cupric metaborate is reduced to cuprous metaborate in a luminous flame.

Statement II : Cuprous metaborate is obtained by heating boric anhydride and copper sulphate in a non-luminous flame.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

Open solution

Q141. JEE Main 2020 (Online) 6th September Evening Slot

Mischmetal is an alloy consisting mainly of :

(A) lanthanoid and actinoid metals

(B) actinoid and transition metals

(C) lanthanoid metals

(D) actinoid metals

Open solution

Q142. JEE Main 2020 (Online) 6th September Morning Slot

The lanthanoid that does NOT show +4 oxidation state is :

(A) Tb

(B) Dy

(C) Ce

(D) Eu

Open solution

Q143. JEE Main 2020 (Online) 6th September Morning Slot

The set that contains atomic numbers of only transition elements, is :

(A) 21, 32, 53, 64

(B) 9, 17, 34, 38

(C) 37, 42, 50, 64

(D) 21, 25, 42, 72

Open solution

Q144. JEE Main 2020 (Online) 5th September Morning Slot

The correct electronic configuration and spin-only magnetic moment (BM) of Gd^{3+} (Z = 64), respectively, are :

(A) $[Xe]5f^7$ and 8.9

(B) $[Xe]4f^7$ and 7.9

(C) $[Xe]5f^7$ and 7.9

(D) $[Xe]4f^7$ and 8.9

Open solution

Q145. JEE Main 2020 (Online) 5th September Morning Slot

In the sixth period, the orbitals that are filled are :

(A) 6s, 5d, 5f, 6p

(B) 6s, 4f, 5d, 6p

(C) 6s, 6p, 6d, 6f

(D) 6s, 5f, 6d, 6p

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q146. JEE Main 2020 (Online) 4th September Evening Slot

The incorrect statement(s) among (a) - (c) is (are)

(a) W(VI) is more stable than Cr(VI).

(b) In the presence of HCl, permanganate titrations provide satisfactory results.

(c) Some lanthanoid oxides can be used as phosphors.

(A) (a) and (b) only **(B)** (a) only **(C)** (b) only **(D)** (b) and (c) only

Open solution

Q147. JEE Main 2020 (Online) 4th September Morning Slot

The elements with atomic numbers 101 and 104 belong to, respectively :

(A) Group 6 and Actinoids **(B)** Actinoids and Group 4

(C) Group 11 and Group 4 **(D)** Actinoids and Group 6

Open solution

Q148. JEE Main 2020 (Online) 3rd September Evening Slot

The incorrect statement is :

(A) In manganate and permanganate ions, the π-bonding takes place by overlap of p-orbitals of oxygen and d-orbitals of manganese

(B) Manganate ion is green in colour and permanganate ion is purple in colour

(C) Manganate and permanganate ions are paramagnetic

(D) Manganate and permanganate ions are tetrahedral

Open solution

Q149. JEE Main 2020 (Online) 9th January Evening Slot

5 g of zinc is treated separately with an excess of

(a) dilute hydrochloric acid and
(b) aqueous sodium hydroxide.

The ratio of the volumes of H_2 evolved in these two reactions is :

(A) 1 : 2 **(B)** 1 : 1 **(C)** 1 : 4 **(D)** 2 : 1

Open solution

Q150. JEE Main 2020 (Online) 9th January Morning Slot

The electronic configurations of bivalent europium and trivalent cerium are
(atomic number : Xe = 54, Ce = 58, Eu = 63)

(A) [Xe] $4f^7$ $6s^2$ and [Xe] $4f^2$ $6s^2$ **(B)** [Xe] $4f^2$ and [Xe] $4f^7$

(C) [Xe] $4f^4$ and [Xe] $4f^9$ **(D)** [Xe] $4f^7$ and [Xe] $4f^1$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q151. JEE Main 2020 (Online) 7th January Morning Slot

The atomic radius of Ag is closed to :

(A) Au (B) Cu (C) Hg (D) Ni

Open solution

Q152. JEE Main 2019 (Online) 12th April Evening Slot

Thermal decomposition of a Mn compound (X) at 513 K results in compound Y, MnO_2 and gaseous product. MnO_2 reacts with NaCl and concentrated H_2SO_4 to give a pungent gas Z. X, Y and Z, respectively, are :

(A) $KMnO_4$, K_2MnO_4 and Cl_2 (B) K_2MnO_4, $KMnO_4$ and SO_2

(C) K_3MnO_4, K_2MnO_4 and Cl_2 (D) K_2MnO_4, $KMnO_4$ and Cl_2

Open solution

Q153. JEE Main 2019 (Online) 12th April Evening Slot

The pair that has similar atomic radii is :

(A) Mo and W (B) Mn and Fe (C) Ti and Hf (D) Sc and Ni

Open solution

Q154. JEE Main 2019 (Online) 10th April Evening Slot

The highest possible oxidation states of uranium and plutonium, respectively are :

(A) 4 and 6 (B) 6 and 4 (C) 7 and 6 (D) 6 and 7

Open solution

Q155. JEE Main 2019 (Online) 10th April Evening Slot

The correct order of the first ionization enthalpies is :

(A) Ti < Mn < Zn < Ni (B) Zn < Ni < Mn < Ti

(C) Mn < Ti < Zn < Ni (D) Ti < Mn < Ni < Zn

Open solution

Q156. JEE Main 2019 (Online) 10th April Morning Slot

Consider the hydrated ions of Ti^{2+}, V^{2+}, Ti^{3+}, and Sc^{3+}. The correct order of their spin-only magnetic moments is :

(A) $Sc^{3+} < Ti^{3+} < V^{2+} < Ti^{2+}$ (B) $Sc^{3+} < Ti^{3+} < Ti^{2+} < V^{2+}$

(C) $Ti^{3+} < Ti^{2+} < Sc^{3+} < V^{2+}$ (D) $V^{2+} < Ti^{2+} < Ti^{3+} < Sc^{3+}$

Open solution

Q157. JEE Main 2019 (Online) 9th April Evening Slot

The maximum number of possible oxidation states of actinoides are shown by :

(A) nobelium (No) and lawrencium (Lr) (B) actinium (Ac) and thorium (Th)

(C) neptunium (Np) and plutonium (Pu) (D) berkelium (Bk) and californium (Cf)

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q158. JEE Main 2019 (Online) 8th April Evening Slot

The statement that is INCORRECT about the interstitial compounds is :

(A) They are chemically reactive
(B) They have metallic conductivity
(C) They have high melting points
(D) They are very hard

Open solution

Q159. JEE Main 2019 (Online) 8th April Morning Slot

The lanthanide ion that would show colour is :

(A) Sm^{3+} (B) Gd^{3+} (C) Lu^{3+} (D) La^{3+}

Open solution

Q160. JEE Main 2019 (Online) 12th January Evening Slot

The correct order of atomic radii is :

(A) Ho > N > Eu > Ce (B) Ce > Eu > Ho > N (C) N > Ce > Eu > Ho (D) Eu > Ce > Ho > N

Open solution

Q161. JEE Main 2019 (Online) 11th January Evening Slot

$$\underline{A} \xrightarrow{4KOH, O_2} \underset{(Green)}{2\underline{B}} + 2H_2O$$

$$3\underline{B} \xrightarrow{4HCl} \underset{(Purple)}{2\underline{C}} + MnO_2 + 2H_2O$$

$$2\underline{C} \xrightarrow{H_2O.KI} \underset{(Purple)}{2\underline{A}} + 2KOH + \underline{D}$$

In the above sequence of reactions, $\underline{A}$ and $\underline{D}$, respectively, are :

(A) KI and $KMnO_4$ (B) KI and K_2MnO_4 (C) KIO_3 and MnO_2 (D) MnO_2 and KIO_3

Open solution

Q162. JEE Main 2019 (Online) 11th January Morning Slot

The element that usually does NOT show variable oxidation states is :

(A) V (B) Cu (C) Sc (D) Ti

Open solution

Q163. JEE Main 2019 (Online) 10th January Evening Slot

The 71st electron of an element X with an atomic number of 71 enters into the orbital :

(A) 4f (B) 6s (C) 6p (D) 5d

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q164. JEE Main 2019 (Online) 10th January Morning Slot

The effect of lanthanoid contraction in the lanthanoid series of elements by and large means :

(A) increase in atomic radii and decrease in ionic radii

(B) increase in both atomic and ionic radii

(C) decrease in both atomic and ionic radii

(D) decrease in atomic radii and increase in ionic radii

Open solution

Q165. JEE Main 2019 (Online) 9th January Evening Slot

The transition element that has lowest enthalpy of atomisation, is :

(A) Fe (B) Cu (C) V (D) Zn

Open solution

Q166. JEE Main 2018 (Online) 16th April Morning Slot

The **incorrect** statement is :

(A) Cu^{2+} salts give red coloured borax bead test in reducing flame.

(B) Cu^{2+} and Ni^{2+} ions give black precipitate with H_2S in presence of HCl solution.

(C) Ferric ion gives blood red color with potassium thiocyanate.

(D) Cu^{2+} ion gives chocolate coloured precipitate with potassium ferrocyanide solution.

Open solution

Q167. JEE Main 2018 (Online) 16th April Morning Slot

When XO_2 is fused with an alkali metal hydroxide in presence of an oxidizing agent such as KNO_3 ; a dark green product is formed which disproportionates in acidic solution to afford a dark purple solution. X is :

(A) Ti (B) V (C) Cr (D) Mn

Open solution

Q168. JEE Main 2017 (Online) 9th April Morning Slot

Which of the following ions does not liberate hydrogen gas on reaction with dilute acids ?

(A) Ti^{2+} (B) V^{2+} (C) Cr^{2+} (D) Mn^{2+}

Open solution

Q169. JEE Main 2017 (Online) 8th April Morning Slot

The pair of compounds having metals in their highest oxidation state is :

(A) MnO_2 and CrO_2Cl_2

(B) $[NiCl_4]^{2-}$ and $[CoCl_4]^{2-}$

(C) $[Fe(CN)_6]^{3-}$ and $[Cu(CN)_4]^{2-}$

(D) $[FeCl_4]^{-}$ and Co_2O_3

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q170. JEE Main 2017 (Offline)

In the following reactions, ZnO is respectively acting as a/an :

(a) $ZnO + Na_2O \rightarrow Na_2ZnO_2$

(b) $ZnO + CO_2 \rightarrow ZnCO_3$

(A) base and base (B) acid and acid (C) acid and base (D) base and acid

Open solution

Q171. JEE Main 2016 (Online) 10th April Morning Slot

The transition metal ions responsible for color in ruby and emerald are, respectively :

(A) Cr^{3+} and Co^{3+} (B) Co^{3+} and Cr^{3+}

(C) Co^{3+} and Co^{3+} (D) Cr^{3+} and Cr^{3+}

Open solution

Q172. JEE Main 2016 (Online) 9th April Morning Slot

Which one of the following species is stable in aqueous solution?

(A) Cr^{2+} (B) Cu^{+} (C) MnO_4^{3-} (D) MnO_4^{2-}

Open solution

Q173. JEE Main 2016 (Offline)

Which of the following compounds is metallic and ferromagnetic ?

(A) CrO_2 (B) VO_2 (C) MnO_2 (D) TiO_2

Open solution

Q174. JEE Main 2015 (Online) 11th April Morning Slot

Which of the following statement is **false** ?

(A) CrO_4^{2-} is tetrahedral in shape

(B) $Cr_2O_7^{2-}$ has a Cr−O−Cr bond

(C) $Na_2\ Cr_2\ O_7$ is a primary standard in volumetry

(D) $Na_2Cr_2O_7$ is less soluble than $K_2Cr_2O_7$

Open solution

Q175. JEE Main 2015 (Offline)

The colour of $KMnO_4$ is due to :

(A) d − d transition (B) L → M charge transfer transition

(C) $\sigma - \sigma^*$ transition (D) M → L charge transfer transition

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q176. JEE Main 2015 (Offline)

Match the catalysts to the correct processes :

Open solution

	Catalyst		Process
(A)	$TiCl_3$	(i)	Wacker process
(B)	$PdCl_2$	(ii)	Ziegler - Natta polymerization
(C)	$CuCl_2$	(iii)	Contact process
(D)	V_2O_5	(iv)	Deacon's process

(A) $(A)-(ii),\ (B)-(iii),\ (C)-(iv),\ (D)-(i)$

(B) $(A)-(iii),\ (B)-(i),\ (C)-(ii),\ (D)-(iv)$

(C) $(A)-(iii),\ (B)-(ii),\ (C)-(iv),\ (D)-(i)$

(D) $(A)-(ii),\ (B)-(i),\ (C)-(iv),\ (D)-(iii)$

Q177. JEE Main 2014 (Online) 12th April Morning Slot

Which one of the following exhibits the largest number of oxidation states ?

Open solution

(A) Ti (22) **(B)** V (23) **(C)** Cr (24) **(D)** Mn (25)

Q178. JEE Main 2014 (Online) 9th April Morning Slot

Chloro compound of Vanadium has only spin magnetic moment of 1.73 BM. This Vanadium chloride has the formula : (at. no. of V = 23)

Open solution

(A) VCl_2 **(B)** VCl_4 **(C)** VCl_3 **(D)** VCl_5

Q179. JEE Main 2014 (Online) 9th April Morning Slot

Which of the following is **not** formed when H_2S reacts with acidic $K_2Cr_2O_7$ solution ?

Open solution

(A) $CrSO_4$ **(B)** $Cr_2(SO_4)_3$ **(C)** K_2SO_4 **(D)** S

Q180. JEE Main 2014 (Offline)

Which series of reactions correctly represents chemical relations related to iron and its compound?

Open solution

(A) $Fe \xrightarrow{dil.\ H_2SO_4} FeSO_4 \xrightarrow{H_2SO_4, O_2} Fe_2(SO_4)_3 \xrightarrow{heat} Fe$

(B) $Fe \xrightarrow{O_2,\ heat} FeO \xrightarrow{dil.\ H_2SO_4} FeSO_4 \xrightarrow{heat} Fe$

(C) $Fe \xrightarrow{Cl_2,\ heat} FeCl_3 \xrightarrow{heat,\ air} FeCl_2 \xrightarrow{Zn} Fe$

(D) $Fe \xrightarrow{O_2,\ heat} Fe_3O_4 \xrightarrow{CO,\ 600^oC} FeO \xrightarrow{CO,\ 700^oC} Fe$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q181. JEE Main 2013 (Online) 23th April Morning Slot

When a small amount of $KMnO_4$ is added to concentrated H_2SO_4, a green oily compound is obtained which is highly explosive in nature. Compound may be :

(A) $MnSO_4$ (B) Mn_2O_7 (C) MnO_2 (D) Mn_2O_3

Open solution

Q182. JEE Main 2013 (Online) 23th April Morning Slot

Identify the incorrect statement :

(A) Cu_2O is colourless.

(B) Copper (I) compounds are colourless except when colour results from charge transfer.

(C) Copper (I) compounds are diamagnetic.

(D) Cu_2S is black.

Open solution

Q183. JEE Main 2013 (Online) 9th April Morning Slot

The element with which of the following outer electron configuration may exhibit the largest number of oxidation states in its compounds :

(A) $3d^5 4s^2$ (B) $3d^8 4s^2$ (C) $3d^7 4s^2$ (D) $3d^6 4s^2$

Open solution

Q184. JEE Main 2013 (Offline)

Which of the following arrangements does not represent the correct order of the property stated against it?

(A) $Ni^{2+} < Co^{2+} < Fe^{2+} < Mn^{2+}$: ionic size

(B) $Co^{3+} < Fe^{3+} < Cr^{3+} < Sc^{3+}$: stability in aqueous solution

(C) Sc < Ti < Cr < Mn : number of oxidation states

(D) $V^{2+} < Cr^{2+} < Mn^{2+} < Fe^{2+}$: paramagnetic behaviour

Open solution

Q185. JEE Main 2013 (Offline)

Four successive members of the first row transition elements are listed below with atomic numbers. Which one of them is expected to have the highest $E^0_{M^{3+}/M^{2+}}$ value?

(A) Mn (Z = 25) (B) Fe (Z = 26) (C) Co (Z = 27) (D) Cr (Z = 24)

Open solution

Q186. AIEEE 2012 (Online) 26th May

Which of the following forms stable $+4$ oxidation state?

(A) $\mathrm{La}(Z = 57)$ (B) $\mathrm{Eu}(Z = 63)$ (C) $\mathrm{Ce}(Z = 58)$ (D) $\mathrm{Gd}(Z = 64)$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q187. AIEEE 2012 (Online) 26th May

The number of unpaired electrons in Gadolinium $[Z = 64]$ is

(A) 3 (B) 8 (C) 6 (D) 2

Open solution

Q188. AIEEE 2012 (Online) 19th May

Which of the following paramagnetic ions would exhibit a magnetic moment (spin only) of the order of $5\ \mathrm{BM}$?
(At. Nos. $\mathrm{Mn} = 25, \mathrm{Cr} = 24, \mathrm{V} = 23, \mathrm{Ti} = 22$)

(A) Mn^{2+} (B) Ti^{2+} (C) V^{2+} (D) Cr^{2+}

Open solution

Q189. AIEEE 2012 (Online) 12th May

Magnetic moment of Gd^{3+} ion $(Z = 64)$ is

(A) $3.62\ \mathrm{BM}$ (B) $9.72\ \mathrm{BM}$ (C) $7.9\ \mathrm{BM}$ (D) $10.60\ \mathrm{BM}$

Open solution

Q190. AIEEE 2012 (Online) 7th May

Which is not the correct Statement? (At. nos. $\mathrm{Ce} = 58, \mathrm{Lu} = 71, \mathrm{La} = 57, \mathrm{Yb} = 70$)

(A) Colour of Yb^{3+} ion is pink.
(B) La^{3+} is diamagnetic.
(C) Ce^{4+} has f^0 configuration.
(D) Lu^{3+} has f^{14} configuration.

Open solution

Q191. AIEEE 2012

Iron exhibits + 2 and +3 oxidation states. Which of the following statements about iron is incorrect?

(A) Ferrous oxide is more basic in nature than the ferric oxide
(B) Ferrous compounds are relatively more ionic than the corresponding ferric compounds
(C) Ferrous compounds are less volatile than the corresponding ferric compounds
(D) Ferrous compounds are more easily hydrolysed than the corresponding ferric compounds.

Open solution

Q192. AIEEE 2011

In context of the lanthanoids, which of the following statements is not correct?

(A) All the members exhibit +3 oxidation state
(B) Because of similar properties the separation of lanthanoids is not easy.
(C) Availability of 4f electrons results in the formation of compounds in +4 state for all the members of the series.
(D) There is a gradual decrease in the radii of the members with increasing atomic number in the series.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q193. AIEEE 2011

Outer electronic configuration of Gd (Atomic no : 64) is -

(A) $4f^8 5d^0 6s^2$ (B) $4f^4 5d^4 6s^2$ (C) $4f^7 5d^1 6s^2$ (D) $4f^3 5d^5 6s^2$

Open solution

Q194. AIEEE 2010

The correct order of $E^o_{M^{2+}/M}$ values with negative sign for the four successive elements Cr, Mn, Fe and Co is :

(A) Mn > Cr > Fe > Co
(B) Cr > Fe > Mn > Co
(C) Fe > Mn > Cr > Co
(D) Cr > Mn > Fe > Co

Open solution

Q195. AIEEE 2009

In context with the transition elements, which of the following statements is incorrect ?

(A) In addition to the normal oxidation states, the zero oxidation state is also shown by these elements in complexes.
(B) In the highest oxidation states, the transition metal show basic character and form cationic complexes.
(C) In the highest oxidation states of the first five transition elements (Sc to Mn), all the 4s and 3d electrons are used for bonding
(D) Once the d^5 configuration is exceeded, the tendency to involve all the 3d electrons in bonding decreases.

Open solution

Q196. AIEEE 2009

Knowing that the Chemistry of lanthanoids (Ln) is dominated by its +3 oxidation state, which of the following statements is incorrect?

(A) Because of the large size of the Ln (III) ions the bonding in its compounds is predominantly ionic in character.
(B) The ionic sizes of Ln (III) decrease in general with increasing atomic number.
(C) Ln (III) compounds are generally colourless
(D) Ln (III) hydroxides are mainly basic in character.

Open solution

Q197. AIEEE 2008

Larger number of oxidation states are exhibited by the actinoids than those by the lanthanoids, the main reason being :

(A) 4f orbitals more diffused than the 5f orbitals
(B) lesser energy difference between 5f and 6d than between 4f and 5d orbitals
(C) more energy difference between 5f and 6d than between 4f and 5d orbitals
(D) more reactive nature of the actinoids than the lanthanoids

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q198. AIEEE 2008

Amount of oxalic acid present in a solution can be determined by its titration with $KMnO_4$ solution in the presence of H_2SO_4. The titration gives unsatisfactory result when carried out in the presence of HCl, because HCl :

(A) gets oxidised by oxalic acid to chlorine

(B) furnishes H^+ ions in addition to those from oxalic acid

(C) reduces permanganate to Mn^{2+}

(D) oxaidises oxalic acid to carbon doxide and water

Open solution

Q199. AIEEE 2007

Identify the incorrect statement among the following :

(A) d-Block elements show irregular and erratic chemical properties among themselves

(B) La and Lu have partially filled d orbitals and no other partially filled orbitals

(C) The chemistry of various lanthanoids is very similar

(D) 4f and 5f orbitals are equally shielded

Open solution

Q200. AIEEE 2007

The actinoids exhibits more number of oxidation states in general than the lanthanoids. This is because :

(A) the 5f orbitals are more buried than the 4f orbitals

(B) there is a similarity between 4f and 5f orbitals in their angular part of the wave function

(C) the actinoids are more reactive than the lanthanoids

(D) the 5f orbitals extend further from the nucleus than the 4f orbitals

Open solution

Q201. AIEEE 2006

The "spin-only" magnetic moment [in units of Bohr magneton, (μ_B)] of Ni^{2+} in aqueous solution would be : (Atomic number of Ni = 28)

(A) 2.84 (B) 4.90 (C) 0 (D) 1.73

Open solution

Q202. AIEEE 2006

Lanthanoid contraction is caused due to :

(A) the appreciable shielding on outer electrons by 4f electrons from the nuclear charge

(B) the appreciable shielding on outer electrons by 5d electrons from the nuclear charge

(C) the same effective nuclear charge from Ce to Lu

(D) the imperfect shielding on outer electrons by 4f electrons from the nuclear charge

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q203. AIEEE 2006

A metal, M forms chlorides in its $+2$ and $+4$ oxidation states. Which of the following statements about these chlorides is correct?

(A) MCl_2 is more ionic than MCl_4

(B) MCl_2 is more easily hydrolysed than MCl_4

(C) MCl_2 is more volatile than MCl_4

(D) MCl_2 is more soluble in anhydrous ethanol than MCl_4.

Open solution

Q204. AIEEE 2005

The lanthanide contraction is responsible for the fact that :

(A) Zr and Y have about the same radius

(B) Zr and Nb have similar oxidation state

(C) Zr and Hf have about the same radius

(D) Zr and Zn have the same oxidation

Open solution

Q205. AIEEE 2005

Which of the following factors may be regarded as the main cause of lanthanide contraction?

(A) Poor shielding of one of 4f electron by another in the subshell

(B) Effective shielding of one of 4f electrons by another in the subshell

(C) Poorer shielding of 5d electrons by 4f electrons

(D) Greater shielding of 5d electrons by 4f electrons

Open solution

Q206. AIEEE 2005

The oxidation state of chromium in the final product formed the reaction between KI and acidified potassium dichromate solution is :

(A) $+3$ (B) $+2$ (C) $+6$ (D) $+4$

Open solution

Q207. AIEEE 2005

Calomel (Hg_2Cl_2) on reaction with ammonium hydroxide gives :

(A) HgO

(B) Hg_2O

(C) $NH_2 - Hg - Hg - Cl$

(D) $HgNH_2Cl$

Open solution

Q208. AIEEE 2004

Excess of KI reacts with $CuSO_4$ solution and then $Na_2S_2O_3$ solution is added to it. Which of the statements is incorrect for this reaction?

(A) Cu_2I_2 is reduced

(B) Evolved I_2 is reduced

(C) $Na_2S_2O_3$ is oxidized

(D) CuI_2 is formed

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q209. AIEEE 2004

Cerium (Z = 58) is an important member of the lanthanoids. Which of the following statements about cerium is incorrect?

(A) The common oxidation states of cerium are +3 and +4

(B) Cerium (IV) acts as an oxidizing agent

(C) The +4 oxidation state of cerium is not known in solutions

(D) The +3 oxidation state of cerium is more stable than the +4 oxidation state

Open solution

Q210. AIEEE 2004

Of the following outer electronic configurations of atoms, the highest oxidation state is achieved by which one of them?

(A) $(n-1)d^3ns^2$

(B) $(n-1)d^5ns^1$

(C) $(n-1)d^8ns^2$

(D) $(n-1)d^5ns^2$

Open solution

Q211. AIEEE 2003

The number of d-electrons retained in Fe^{2+} (At no of Fe = 26) ion is :

(A) 4 (B) 5 (C) 6 (D) 3

Open solution

Q212. AIEEE 2003

For making good quality mirrors, plates of float glass are used. These are obtained by floating molten glass over a liquid metal which does not solidify before glass. The metal used can be :

(A) tin (B) sodium (C) magnesium (D) mercury

Open solution

Q213. AIEEE 2003

What would happen when a solution of potassium chromate is treated with an excess of dilute nitric acid?

(A) $Cr_2O_7^{2-}$ and H_2O are formed

(B) CrO_4^{2-} is reduced to +3 state of Cr

(C) CrO_4^{2-} is oxidized to +7 state of Cr

(D) Cr^{3+} and $Cr_2O_7^{2-}$ are formed

Open solution

Q214. AIEEE 2003

Which one of the following nitrates will leave behind a metal on strong heating?

(A) Copper nitrate

(B) Manganese nitrate

(C) Silver nitrate

(D) Ferric nitrate

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q215. AIEEE 2003

The radius of La^{3+} (Atomic number of La = 57) is 1.06 Å. Which one of the following given values will be closest to the radius of Lu^{3+} (Atomic number of Lu = 71) ?

(A) 1.60 Å (B) 1.40 Å (C) 0.85 Å (D) 1.60 Å

Open solution

Q216. AIEEE 2003

Which one of the following statements is correct?

(A) From a mixed precipitate of AgCl and AgI, ammonia solution dissolves only AgCl

(B) Ferric ions give a deep green precipitate on adding potassium ferrocyanide solution

(C) On boiling a solution having K^+, Ca^{2+} and HCO_3 ions we get a precipitate of $K_2Ca(CO_3)_2$.

(D) Manganese salts give a violet borax bead test in the reducing flame

Open solution

Q217. AIEEE 2003

A red solid is insoluble in water. However it becomes soluble if some KI is added to water. Heating the red solid in a test tube results in liberation of some violet colored fumes and droplets of a metal appear on the cooler parts of the test tube. The red solid is :

(A) HgI_2 (B) HgO

(C) Pb_3O_4 (D) $(NH_4)_2Cr_2O_7$

Open solution

Q218. AIEEE 2002

When $KMnO_4$ acts as an oxidising agent and ultimately forms $[MnO_4]^{-2}$, MnO_2, Mn_2O_3, Mn^{+2} then the number of electrons transferred in each case respectively is :

(A) 4, 3, 1, 5 (B) 1, 5, 3, 7 (C) 1, 3, 4, 5 (D) 3, 5, 7, 1

Open solution

Q219. AIEEE 2002

Which of the following ions has the maximum magnetic moment?

(A) Mn^{2+} (B) Fe^{2+} (C) Ti^{2+} (D) Cr^{2+}

Open solution

Q220. AIEEE 2002

Most common oxidation states of Ce (cerium) are :

(A) +3, +4 (B) +2, +3 (C) +2, +4 (D) +3, +5

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Numerical 41 questions

Q221. JEE Main 2026 (Online) 2nd April Evening Shift

Number of paramagnetic ions among the following d- and f-block metal ions is _____.

Mn^{2+}, Cu^{2+}, Zn^{2+}, Yb^{2+}, Sc^{3+}, La^{3+}, Gd^{3+}, Lu^{3+}, Ti^{4+}, Ce^{4+}

(Atomic number of Mn = 25, Cu = 29, Zn = 30, Yb = 70, Sc = 21, La = 57, Gd = 64, Lu = 71, Ti = 22, Ce = 58)

Open solution

Q222. JEE Main 2026 (Online) 22nd January Evening Shift

Among the following oxides of $3d$ elements, the number of mixed oxides are ____ .

$Ti_2O_3,\ V_2O_4, Cr_2O_3, Mn_3O_4, Fe_3O_4, Fe_2O_3, Co_3O_4$

Open solution

Q223. JEE Main 2026 (Online) 21st January Morning Shift

Consider the following reactions :

$$NaCl + K_2Cr_2O_7 + H_2SO_4 \rightarrow A + KHSO_4 + NaHSO_4 + H_2O$$
$$A + NaOH \rightarrow B + NaCl + H_2O$$
$$B + H_2SO_4 + H_2O_2 \rightarrow C + Na_2SO_4 + H_2O$$

In the product ' C', ' X ' is the number of O_2^{2-} units, ' Y ' is the total number of oxygen atoms present and ' Z ' is the oxidation state of Cr. The value of $X + Y + Z$ is ____ .

Open solution

Q224. JEE Main 2025 (Online) 3rd April Evening Shift

Among, Sc, Mn, Co and Cu , identify the element with highest enthalpy of atomisation. The spin only magnetic moment value of that element in its +2 oxidation state is_____BM (in nearest integer).

Open solution

Q225. JEE Main 2025 (Online) 3rd April Morning Shift

Consider the following reactions

$$A + \underset{\text{amount}}{\overset{\text{Little}}{NaCl}} + H_2SO_4 \rightarrow CrO_2Cl_2 + \text{ Side Products}$$
$$CrO_2Cl_{2\,(\text{Vapour})} + NaOH \rightarrow B + NaCl + H_2O$$
$$B + H^+ \rightarrow C + H_2O$$

The number of terminal ' O ' present in the compound ' C ' is_____

Open solution

Q226. JEE Main 2025 (Online) 29th January Morning Shift

The molar mass of the water insoluble product formed from the fusion of chromite ore ($FeCr_2O_4$) with Na_2CO_3 in presence of O_2 is _____ g mol^{-1}.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q227. JEE Main 2025 (Online) 22nd January Evening Shift

Niobium (Nb) and ruthenium (Ru) have " x " and " y " number of electrons in their respective 4 d orbitals. The value of $x+y$ is _____.

Open solution

Q228. JEE Main 2024 (Online) 9th April Evening Shift

A transition metal 'M' among $\mathrm{Sc}, \mathrm{Ti}, \mathrm{V}, \mathrm{Cr}, \mathrm{Mn}$ and Fe has the highest second ionisation enthalpy. The spin-only magnetic moment value of M^{+} ion is _____ BM (Nearest integer)

(Given atomic number $\mathrm{Sc}: 21, \mathrm{Ti}: 22, \mathrm{V}: 23, \mathrm{Cr}: 24, \mathrm{Mn}: 25, \mathrm{Fe}: 26$)

Open solution

Q229. JEE Main 2024 (Online) 9th April Morning Shift

Number of colourless lanthanoid ions among the following is _____.

$\mathrm{Eu}^{3+}, \mathrm{Lu}^{3+}, \mathrm{Nd}^{3+}, \mathrm{La}^{3+}, \mathrm{Sm}^{3+}$

Open solution

Q230. JEE Main 2024 (Online) 6th April Evening Shift

Among $\mathrm{VO}_2^{+}, \mathrm{MnO}_4^{-}$ and $\mathrm{Cr}_2\mathrm{O}_7^{2-}$, the spin-only magnetic moment value of the species with least oxidising ability is _____ BM (Nearest integer).

(Given atomic member $\mathrm{V}=23, \mathrm{Mn}=25, \mathrm{Cr}=24$)

Open solution

Q231. JEE Main 2024 (Online) 6th April Morning Shift

Among $\mathrm{CrO}, \mathrm{Cr}_2\mathrm{O}_3$ and CrO_3, the sum of spin-only magnetic moment values of basic and amphoteric oxides is _____ $10^{-2}\mathrm{BM}$ (nearest integer).

(Given atomic number of Cr is 24)

Open solution

Q232. JEE Main 2024 (Online) 5th April Evening Shift

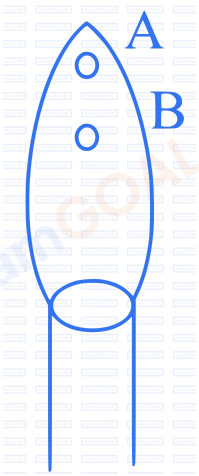
The fusion of chromite ore with sodium carbonate in the presence of air leads to the formation of products A and B along with the evolution of CO_2. The sum of spin-only magnetic moment values of A and B is _____ B.M. (Nearest integer)

[Given atomic number : $\mathrm{C}: 6, \mathrm{Na}: 11, \mathrm{O}: 8, \mathrm{Fe}: 26, \mathrm{Cr}: 24$]

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q233. JEE Main 2024 (Online) 5th April Morning Shift



Open solution

In a borax bead test under hot condition, a metal salt (one from the given) is heated at point B of the flame, resulted in green colour salt bead. The spin-only magnetic moment value of the salt is _____ BM (Nearest integer) [Given atomic number of $\mathrm{Cu} = 29, \mathrm{Ni} = 28, \mathrm{Mn} = 25, \mathrm{Fe} = 26$]

Q234. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

A first row transition metal with highest enthalpy of atomisation, upon reaction with oxygen at high temperature forms oxides of formula $\mathrm{M_2O_n}$ (where $\mathrm{n} = 3, 4, 5$). The 'spin-only' magnetic moment value of the amphoteric oxide from the above oxides is _____ BM (near integer)

(Given atomic number:
$\mathrm{Sc} : 21, \mathrm{Ti} : 22, \ \mathrm{V} : 23, \mathrm{Cr} : 24, \mathrm{Mn} : 25, \mathrm{Fe} : 26, \mathrm{Co} : 27, \mathrm{Ni} : 28, \mathrm{Cu} : 29, \mathrm{Zn} : 30$)

Q235. JEE Main 2024 (Online) 4th April Morning Shift

Open solution

Consider the following reaction

$\mathrm{MnO_2 + KOH + O_2 \rightarrow A + H_2O}$.

Product 'A' in neutral or acidic medium disproportionate to give products 'B' and 'C' along with water. The sum of spin-only magnetic moment values of B and C is _____ BM. (nearest integer)
(Given atomic number of Mn is 25)

Q236. JEE Main 2024 (Online) 27th January Evening Shift

Open solution

Total number of ions from the following with noble gas configuration is _____.

$\mathrm{Sr}^{2+}(z = 38), \mathrm{Cs}^{+}(z = 55), \mathrm{La}^{2+}(z = 57), \mathrm{Pb}^{2+}(z = 82), \mathrm{Yb}^{2+}(z = 70)$ and $\mathrm{Fe}^{2+}(z = 26)$

Q237. JEE Main 2023 (Online) 15th April Morning Shift

Open solution

In Chromyl chloride, the oxidation state of chromium is $(+)$ _____.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q238. JEE Main 2023 (Online) 15th April Morning Shift

The total change in the oxidation state of manganese involved in the reaction of $KMnO_4$ and potassium iodide in the acidic medium is _____.

Open solution

Q239. JEE Main 2023 (Online) 30th January Morning Shift

The number of electrons involved in the reduction of permanganate to manganese dioxide in acidic medium is _____.

Open solution

Q240. JEE Main 2023 (Online) 25th January Morning Shift

How many of the following metal ions have similar value of spin only magnetic moment in gaseous state? _____

(Given : Atomic number V, 23; Cr, 24; Fe, 26; Ni, 28)

V^{3+}, Cr^{3+}, Fe^{2+}, Ni^{3+}

Open solution

Q241. JEE Main 2022 (Online) 30th July Evening Shift

The number of coloured and paramagnetic ions from the following in aqueous solution are _____.

Ti^{3+}, Co^{2+}, Ni^{2+}, Cu^{2+}, Cu^{+}, Ti^{4+}, Zn^{2+}, Sc^{3+}

Given Atomic no.

Sc, 21; Ti, 22; Co, 27; Ni, 28; Cu, 29; Zn, 30

Open solution

Q242. JEE Main 2022 (Online) 28th July Morning Shift

The disproportionation of MnO_4^{2-} in acidic medium resulted in the formation of two manganese compounds A and B. If the oxidation state of Mn in B is smaller than that of A, then the spin-only magnetic moment (μ) value of B in BM is _____. (Nearest integer)

Open solution

Q243. JEE Main 2022 (Online) 26th July Evening Shift

The spin-only magnetic moment value of the compound with strongest oxidizing ability among MnF_4, MnF_3 and MnF_2 is _____ B.M. [nearest integer]

Open solution

Q244. JEE Main 2022 (Online) 25th July Evening Shift

The spin-only magnetic moment value of M^{3+} ion (in gaseous state) from the pairs Cr^{3+} / Cr^{2+}, Mn^{3+} / Mn^{2+}, Fe^{3+} / Fe^{2+} and Co^{3+} / Co^{2+} that has negative standard electrode potential, is _____ B.M. [Nearest integer]

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q245. JEE Main 2022 (Online) 25th July Morning Shift

Among Co^{3+}, Ti^{2+}, V^{2+} and Cr^{2+} ions, one if used as a reagent cannot liberate H_2 from dilute mineral acid solution, its spin-only magnetic moment in gaseous state is _____ B.M. (Nearest integer)

Open solution

Q246. JEE Main 2022 (Online) 29th June Evening Shift

Spin only magnetic moment of $[MnBr_6]^{4-}$ is _____ B.M. (round off to the closest integer)

Open solution

Q247. JEE Main 2022 (Online) 29th June Morning Shift

The number of terminal oxygen atoms present in the product B obtained from the following reaction is _____.

$FeCr_2O_4 + Na_2CO_3 + O_2 \rightarrow A + Fe_2O_3 + CO_2$

$A + H^+ \rightarrow B + H_2O + Na^+$

Open solution

Q248. JEE Main 2022 (Online) 29th June Morning Shift

An acidified manganate solution undergoes disproportionation reaction. The spin-only magnetic moment value of the product having manganese in higher oxidation state is _____ B.M. (Nearest integer)

Open solution

Q249. JEE Main 2022 (Online) 27th June Morning Shift

The number of statements correct from the following for Copper (at. no. 29) is/are _____.

(A) Cu(II) complexes are always paramagnetic.

(B) Cu(I) complexes are generally colourless

(C) Cu(I) is easily oxidized

(D) In Fehling solution, the active reagent has Cu(I)

Open solution

Q250. JEE Main 2022 (Online) 26th June Morning Shift

The spin-only magnetic moment value of the most basic oxide of vanadium among V_2O_3, V_2O_4 and V_2O_5 is _____ B.M. (Nearest integer)

Open solution

Q251. JEE Main 2022 (Online) 24th June Evening Shift

Manganese (VI) has ability to disproportionate in acidic solution. The difference in oxidation states of two ions it forms in acidic solution is _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q252. JEE Main 2022 (Online) 24th June Morning Shift

The difference in oxidation state of chromium in chromate and dichromate salts is _____.

Open solution

Q253. JEE Main 2021 (Online) 31st August Evening Shift

The value of magnetic quantum number of the outermost electron of Zn^+ ion is _____.

Open solution

Q254. JEE Main 2021 (Online) 27th August Morning Shift

The number of f electrons in the ground state electronic configuration of Np (Z = 93) is _____. (Nearest integer)

Open solution

Q255. JEE Main 2021 (Online) 26th August Morning Shift

The number of 4f electrons in the ground state electronic configuration of Gd^{2+} is _____. [Atomic number of Gd = 64]

Open solution

Q256. JEE Main 2021 (Online) 25th July Evening Shift

Number of electrons present in 4f orbital of Ho^{3+} ion is _____. (Given Atomic No. of Ho = 67)

Open solution

Q257. JEE Main 2021 (Online) 22th July Evening Shift

Number of electrons that Vanadium (Z = 23) has in p-orbitals is equal to _____

Open solution

Q258. JEE Main 2021 (Online) 17th March Evening Shift

In the ground state of atomic Fe(Z = 26), the spin-only magnetic moment is _____ $\times 10^{-1}$ BM. (Round off to the Nearest Integer). [Given : $\sqrt{3}$ = 1.73, $\sqrt{2}$ = 1.41]

Open solution

Q259. JEE Main 2021 (Online) 26th February Evening Shift

In mildly alkaline medium, thiosulphate ion is oxidized by MnO_4^- to "A". The oxidation state of sulphur in "A" is _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q260. JEE Main 2020 (Online) 9th January Evening Slot

The sum of the total number of bonds between chromium and oxygen atoms in chromate and dichromate ions is _____.

Open solution

Q261. JEE Main 2020 (Online) 7th January Evening Slot

Consider the following reactions :

$NaCl + K_2Cr_2O_7 + \underset{(conc.)}{H_2SO_4} \rightarrow$ (A) + side products

(A) + NaOH $\rightarrow$ (B) + Side products

$(B) + \underset{(dilute)}{H_2SO_4} + H_2O_2 \rightarrow$ (C) + Side products

The sum of the total number of atoms in one molecule each of (A) and (B) and (C) is

Open solution

MCQ (More than One Correct Answer) 1 questions

Q262. JEE Main 2022 (Online) 28th July Evening Shift

Which of the following pair is not isoelectronic species?

(At. no. Sm, 62; Er, 68; Yb, 70; Lu, 71; Eu, 63; Tb, 65; Tm, 69)

(A) Sm^{2+} and Er^{3+}

(B) Yb^{2+} and Lu^{3+}

(C) Eu^{2+} and Tb^{4+}

(D) Tb^{2+} and Tm^{4+}

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **B**	3. **B**	4. **C**	5. **D**
6. **B**	7. **D**	8. **C**	9. **C**	10. **B**
11. **D**	12. **A**	13. **A**	14. **B**	15. **D**
16. **A**	17. **C**	18. **B**	19. **B**	20. **C**
21. **A**	22. **B**	23. **D**	24. **A**	25. **B**
26. **D**	27. **B**	28. **C**	29. **A**	30. **C**
31. **C**	32. **B**	33. **C**	34. **A**	35. **D**
36. **C**	37. **C**	38. **A**	39. **D**	40. **C**
41. **B**	42. **B**	43. **A**	44. **C**	45. **B**
46. **D**	47. **B**	48. **C**	49. **A**	50. **D**
51. **B**	52. **C**	53. **A**	54. **A**	55. **A**
56. **A**	57. **D**	58. **A**	59. **C**	60. **C**
61. **B**	62. **D**	63. **A**	64. **A**	65. **C**
66. **A**	67. **C**	68. **A**	69. **B**	70. **A**
71. **B**	72. **B**	73. **A**	74. **B**	75. **D**
76. **A**	77. **A**	78. **A**	79. **B**	80. **C**
81. **B**	82. **D**	83. **C**	84. **D**	85. **A**
86. **D**	87. **D**	88. **B**	89. **C**	90. **A**
91. **C**	92. **B**	93. **C**	94. **A**	95. **D**
96. **C**	97. **A**	98. **D**	99. **B**	100. **C**
101. **A**	102. **A**	103. **C**	104. **C**	105. **C**
106. **A**	107. **B**	108. **B**	109. **A**	110. **B**
111. **D**	112. **D**	113. **C**	114. *Scan QR*	115. **B**
116. **B**	117. **B**	118. **D**	119. **D**	120. **A**
121. **B**	122. **A**	123. **C**	124. **A**	125. **C**
126. **B**	127. **B**	128. **D**	129. **A**	130. **C**
131. **C**	132. **D**	133. **C**	134. **C**	135. **D**
136. **A**	137. **D**	138. **C**	139. **C**	140. **C**
141. **C**	142. **D**	143. **D**	144. **B**	145. **B**
146. **C**	147. **B**	148. **C**	149. **B**	150. **D**

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

151. **A**	152. **A**	153. **A**	154. **D**	155. **D**
156. **B**	157. **C**	158. **A**	159. **A**	160. **D**
161. **D**	162. **C**	163. **D**	164. **C**	165. **B**
166. **B**	167. **D**	168. **D**	169. **A**	170. **C**
171. **D**	172. **D**	173. **A**	174. *Scan QR*	175. **B**
176. **D**	177. **D**	178. **B**	179. **A**	180. **D**
181. **B**	182. **A**	183. **A**	184. **D**	185. **C**
186. **C**	187. **B**	188. **D**	189. **C**	190. **A**
191. **D**	192. **C**	193. **C**	194. **A**	195. **B**
196. **C**	197. **B**	198. **C**	199. **D**	200. **D**
201. **A**	202. **D**	203. **A**	204. **C**	205. **A**
206. **A**	207. **D**	208. **D**	209. **C**	210. **D**
211. **C**	212. **D**	213. **A**	214. **C**	215. **C**
216. **A**	217. **A**	218. **C**	219. **A**	220. **A**

Numerical

221. **3**	222. **3**	223. **13**	224. **4**	225. **6**
226. **160**	227. **11**	228. **6**	229. **2**	230. **0**
231. **877**	232. **6**	233. **6**	234. **0**	235. **4**
236. **2**	237. **6**	238. **5**	239. **3**	240. **2**
241. **4**	242. **4**	243. **5**	244. **4**	245. **5**
246. **6**	247. **6**	248. **0**	249. **3**	250. **3**
251. **3**	252. **0**	253. **0**	254. **18**	255. **7**
256. **10**	257. **12**	258. **49**	259. **6**	260. **18**
261. **18**				

MCQ (More than One Correct Answer)

262. **A, D**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs